Standard Operating Procedure for Resolving Routing Protocol Flap Failure on Arista Access Node JAY-ACC-55 at Jayanagar Metro Site

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6901

Description

The routing protocol flap failure on the Arista access node JAY-ACC-55 at the Jayanagar Metro Site is primarily attributed to the dynamic nature of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon is a direct consequence of the complex interplay between the BGP route reflector and the OSPF/ISIS routing protocols, which govern the exchange of routing information between the access node and its neighboring devices. The Arista EOS software, running on the access node, interprets and processes the routing protocol messages, and any anomalies or inconsistencies in these messages can trigger a routing protocol flap, resulting in a loss of OSPF adjacency and ISIS LSP flooding. Furthermore, the high-speed, low-latency fabric provided by the Arista hardware can exacerbate the effects of routing protocol flaps, leading to a cascade of failure events that can impact network stability and availability.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that the power and environmental conditions are within the specified operating parameters.

- Confirm that the redundant links are operational and that the network is not experiencing any other failures or issues.

- Ensure that the Arista EOS software is running the latest version and that all necessary patches and updates have been applied.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version 1, local router ID is 10.10.10.10, Description: This command displays the BGP summary information, including the BGP table version, local router ID, and the number of active paths.
```

```
Command: show ospf database, Expected Output: OSPF database has 10 entries, Description: This command displays the OSPF database information, including the number of entries and the contents of the database.

Command: show isis database, Expected Output: ISIS database has 10 entries, Description: This command displays the ISIS database information, including the number of entries and the contents of the database.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address, Description: This command displays the OSPF neighbor information, including the neighbor ID, priority, state, dead time, and address.
```

Resolution Steps

Step 1: Step: Verify BGP Route Churn, Command: show ip bgp summary, Description: Verify that there are no BGP route churn events occurring on the access node. If BGP route churn is detected, proceed to the next step.

Step 2: Step: Reset OSPF Adjacency, Command: clear ip ospf process, Description: Reset the OSPF process on the access node to re-establish OSPF adjacency.

Step 3: Step: Clear ISIS LSP Flooding, Command: clear isis database, Description: Clear the ISIS LSP flooding on the access node to resolve the issue.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version 1, local router ID is 10.10.10.10, Description: Verify that the BGP table version and local router ID are correct.

Command: show ospf database, Expected Output: OSPF database has 10 entries, Description: Verify that the OSPF database information is correct.

Command: show isis database, Expected Output: ISIS database has 10 entries, Description: Verify that the ISIS database information is correct.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: show running-config, Description: Restore the original configuration on the access node to its previous state.